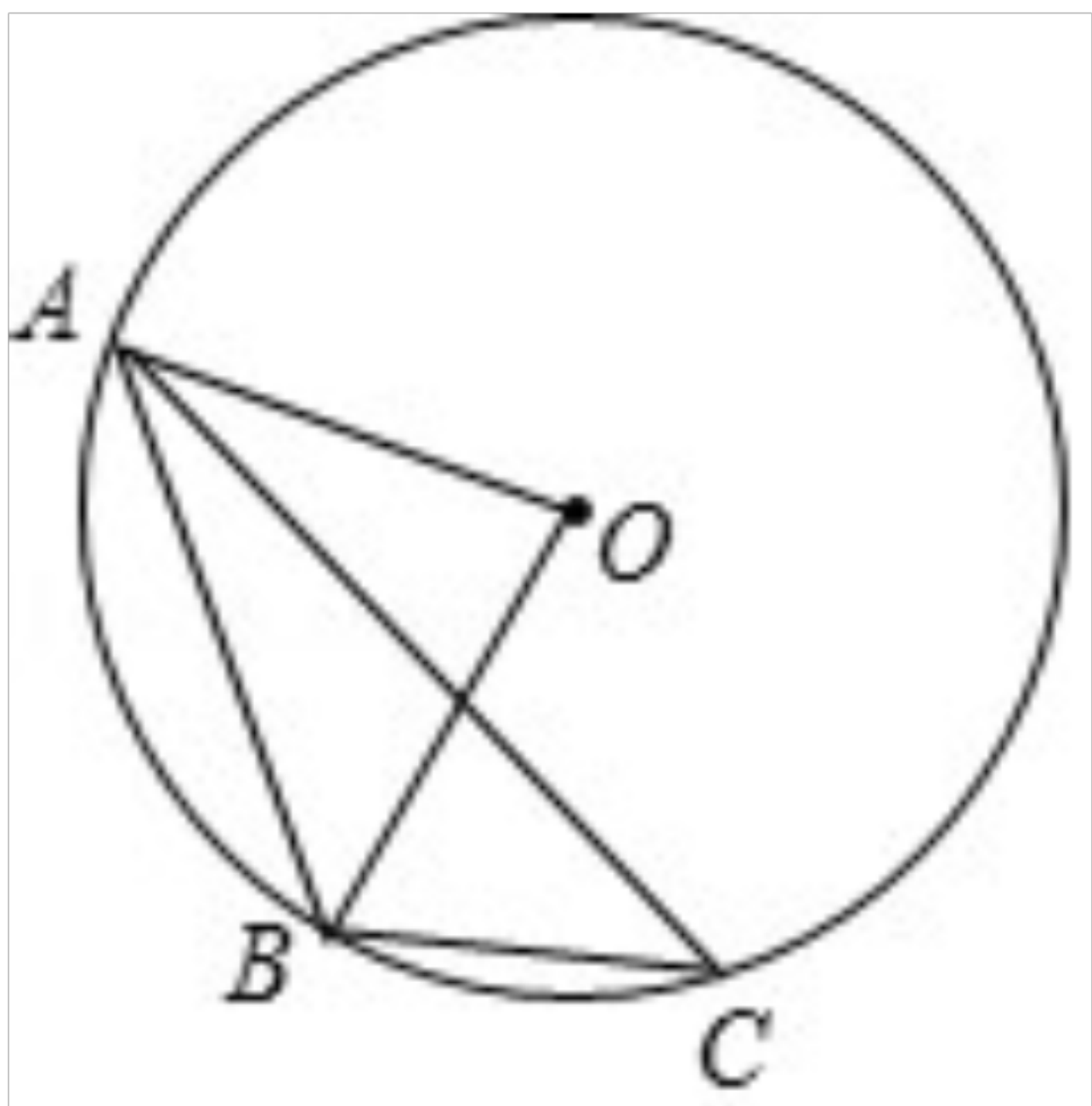




Question:

In the figure, points A, B, and C are on circle O, and $\angle ABO = 50^\circ$. What is the measure of $\angle ACB$?

- A. 40° B. 30° C. 45° D. 50°

Answer: A. 40°



<THINKING> Given that $\angle ABO = 50^\circ$, we need to determine the measure of $\angle ACB$. Since $\angle ABO$ is a central angle, we incorrectly assume that $\angle ACB$ is half of the central angle, leading to:
$$\angle ACB = \frac{1}{2} \times 50^\circ = 25^\circ$$
Since 25° is not an option, we further assume that $\angle ACB$ is the supplementary angle to $\angle ABO$, giving:
$$\angle ACB = 180^\circ - 50^\circ = 130^\circ$$
Since 130° is also not an option, we mistakenly reinterpret $\angle ABO$ as an inscribed angle rather than a central angle, leading to:
$$\angle ACB = 50^\circ$$
</THINKING>

<ANSWER> D. 50° </ANSWER>

<WATCHING>Point:
A: (8.15, 4.68)
B: (-7.61, -6.76)
C: (8.49, -6.76)
Line:
(8.15, 4.68) -- (8.27, 0.68) -- (8.39, -3.32) -- (8.49, -6.76)
(-7.61, -6.76) -- (-4.37, -4.41) -- (-1.14, -2.06) -- (2.1, 0.29) -- (5.34, 2.64) -- (8.15, 4.68)
(-7.61, -6.76) -- (-3.61, -6.76) -- (0.39, -6.76) -- (4.39, -6.76) -- (8.39, -6.76) -- (8.49, -6.76)
</WATCHING>

<THINKING> Since triangle ABC is a right triangle with angle C as the right angle, we use the Pythagorean theorem:
$$AB^2 = AC^2 + BC^2$$
Given:
- AC = 3
- AB = 5
Calculate BC:
$$AB^2 = AC^2 + BC^2$$
$$5^2 = 3^2 + BC^2$$
$$25 = 9 + BC^2$$
$$BC^2 = 16$$
$$BC = 4$$
The sine of angle A is the ratio of the length of the side opposite angle A (which is BC) to the hypotenuse (AB):
$$\sin A = \frac{BC}{AB} = \frac{4}{5}$$
</THINKING>

<ANSWER> C </ANSWER>